
ASCII Interface

The ASCII interface converts data in the SCS binary database to an ASCII format file you can edit with any text editor. The interface also converts an ASCII file (in the appropriate format) to the SCS binary format. You can interface with any CAE/CAD system *via* the SCS ASCII interface (though additional manual editing may be required in some cases).

Once a database has been converted to ASCII format, you can modify it and then translate it back to SCS format. This allows various filtering and processing programs to be used in the SCS environment.

The following ASCII interface conversion programs can be accessed from the SCS Executive Utilities menu:

- ◆ Schem to ASCII
- ◆ ASCII to Schem
- ◆ Symbol to ASCII
- ◆ ASCII to Symbol

ASCII / Symbol Interface

An SCS symbol file contains the information required to describe a symbol and its attributes. The database entries include graphic elements, fixed text, symbol attributes, attribute display windows, pins, and pin attributes.

SCS calculates an “extent” box for each symbol. This box is a rectangle that encloses all the points that define the various elements of the symbol. Unless you choose a different point, the upper-left corner of this box is the default origin for the symbol.

SCS uses a grid system for coordinates. The grid spacing is defined by the Grid Size and Grid Units parameters in the Sheet Layout dialog box of the INI Editor. The Primary grid is divided into a Secondary grid with four points in each major grid interval.

The Secondary grid is used for all coordinates in the symbol database. When the Symbol Editor stores the symbol data base it recalculates the extent box and adjusts the coordinates so that all coordinates are positive integers. The Symbol Editor also limits the size of a symbol to 400 Primary (or 1600 Secondary) grid units in each dimension.

The ASCII to Symbol and Symbol to ASCII conversion utilities are called from the Utilities menu of the SCS Executive.

Interface Format

The interface is a line-oriented ASCII text format. Each element in the symbol is represented as a single line in the text file. Each line consists of:

- ♦ A keyword that begins the line and identifies the type of element described.
- ♦ A list of parameters describing the element following the keyword.

The following conventions apply within each line:

- ♦ Keywords are not case-sensitive.
- ♦ Parameters are delimited by spaces and tabs.
- ♦ Text strings appear at the end of the line and can contain embedded blanks.
- ♦ Coordinate locations are given as pairs of numbers. X coordinates increase to the right and Y coordinates increase to the bottom of the symbol. All coordinate locations and distances are given in Secondary grid units.

In the following description of the format, keywords are shown in roman text and parameters are shown in italics.

version *number*

The SCS symbol database version number is specified on the first line of the file.

symboltype *type*

The symboltype should be the second line of the file. The *type* parameter is BLOCK, CELL, COMPONENT, GATE, GRAPHIC, MASTER, or PIN. If this parameter is not specified, the *type* defaults to BLOCK.

Each of the following formats for graphic elements contains a parameter that specifies line weight. This parameter can have the value NORMAL or WIDE. These keywords can be abbreviated as N or W.

On UNIX systems, the graphic-element format contains a parameter that specifies line style. This parameter can have the value Dash, Dot, or DashDot. If the parameter is missing, the line is drawn solid.

line *width x1 y1 x2 y2 style*

The x-y coordinates are the line's end points.

rectangle *width x1 y1 x2 y2 style*

The x-y coordinates are two opposite corners of the rectangle.

circle width x1 y1 x2 y2 style

The x-y coordinates are the opposite corners of a square, with the square enclosing the circle. If a rectangle is specified, it is converted to a square with the same upper-left corner as the rectangle. The sides of the square are the length of the shorter sides of the rectangle.

arc width x1 y1 x2 y2 x3 y3 x4 y4 style

The first two x-y coordinates describe a circle (see above) containing the arc. The third and fourth coordinate pairs are the starting and ending points of the arc. If these points are not on the circle, the lines joining these points to the circle's center set the boundaries of the arc. The arc is drawn counterclockwise from the starting to the ending point.

text x1 y1 justify font string

The x-y coordinate is the origin for drawing the text.

justify specifies the horizontal (vertical) justification of the text. It can have the value LEFT, RIGHT, CENTER, VLEFT, VRIGHT, or VCENTER. Any value can be abbreviated to just the first two characters. Horizontal (vertical) text is always justified vertically (horizontally) to the center of the character.

font specifies the size of the text and can be 0–7. (In the UNIX / Motif version, there are only three font sizes, 2, 3, and 4. DOS sizes 0–2 are mapped to UNIX size 2, DOS size 3 is mapped to UNIX size 3, and DOS sizes 4–7 are mapped to UNIX size 4.)

string is the text string. The string consists of the first non-blank character after the *font* parameter and all following characters.

symattr name string

name specifies the attribute and must match one of the attribute names listed in the Symbol Attributes section of the INI file.

string is the attribute value assigned, including any attribute modifiers.

window number x1 y1 justify font

number specifies which attribute window is to be drawn.

The x-y coordinate is the origin for drawing the attribute value.

justify specifies the horizontal (vertical) justification of the text. It can have the value LEFT, RIGHT, CENTER, VLEFT, VRIGHT, or VCENTER. Any value can be abbreviated to just the first two characters. Horizontal (vertical) text is always justified vertically (horizontally) to the center of the character.

font specifies the size of the text and can be 0–7. (In the UNIX / Motif version, there are only three font sizes, 0–2. Sizes 4–7 are mapped to UNIX size 2, 3 is mapped to size 1, and 0–2 are mapped to size 0.)

pin x1 y1 justify offset

The *x-y* coordinate is the location of the pin. This location must fall on a Primary grid point.

The *justify* and *offset* parameters indicate where to draw the name of the pin. *offset* is the distance from the pin, and can range from 0 to 31 Secondary grids. *justify* indicates the direction the pin name is offset from the pin and can be LEFT, RIGHT, BOTTOM, TOP, VLEFT, VRIGHT, VBOTTOM, VTOP, or NONE. (Use NONE when you are not giving the pin a name.) Any of these can be abbreviated to just the first two characters.

pinattr name string

Pin attributes apply to the previously defined pin. *name* is the attribute and must match one of the attribute names in the Pin Attributes section of the INI file.

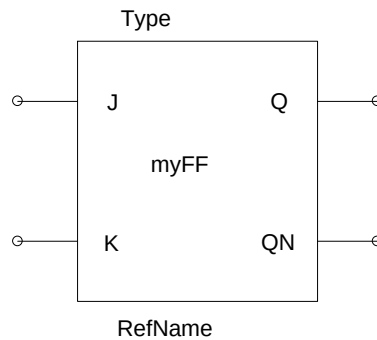
string is the attribute value assigned, including any attribute modifiers.

Example ASCII Symbol File

The following example is the result of translating the symbol in Figure A-1 to the ASCII format.

Figure A-1

Symbol to be Translated to ASCII



```

Version 3
SymbolType COMPONENT
Line N 8 8 0 8
Line N 8 28 0 28
Rectangle N 8 0 40 36
Line N 40 8 48 8
Line N 40 28 48 28
Text 25 17 CENTER 0 myFF
SymAttr SILOSModel .JKFF
Pin 0 8 LEFT 9
PinAttr PinName J
PinAttr Polarity IN
Pin 0 28 LEFT 9
PinAttr PinName K
PinAttr Polarity IN
Pin 48 28 RIGHT 9
PinAttr PinName QN
PinAttr Polarity OUT
Pin 48 8 RIGHT 9
PinAttr PinName Q
PinAttr Polarity OUT
Window 0 24 44 CENTER 0
Window 1 24 -8 CENTER 0

```

ASCII / Schematic Interface

The schematic database is a compact binary file that is created and modified by the Schematic Editor. The ASCII Schematic Interface allows you to translate the SCS schematic database to and from a format that can be modified by a text editor or other program. This allows conversion of schematics from one system to another.

An SCS schematic file contains the information required to describe a schematic including all of its sheets. The database entries include:

- ◆ Wires
- ◆ Buses
- ◆ Net name flags
- ◆ Net attributes
- ◆ Bus taps
- ◆ Symbol instances (both primitive and hierarchical)
- ◆ Symbol attribute overrides
- ◆ Pin attribute overrides
- ◆ Cosmetic graphics
- ◆ Fixed text
- ◆ Tables

A schematic file has one or more *sheets*, each representing a separate page of the printed schematic. Each sheet is divided into grids.

The grid spacing is defined by the Grid Size and Grid Units parameters in the Sheet Layout dialog box of the INI Editor. The Primary grid is divided into a Secondary grid with four points in each major grid interval. All connectivity elements (wires, symbols, pins, and attributes) must lie on the Primary grid. Cosmetic graphics and fixed text can lie on either the Primary or Secondary grid.

The Secondary grid is used for all coordinates in the schematic database. However, all the connectivity elements have coordinates that are integral multiples of four, which forces them to fall on the Primary grid.

The size of a sheet in a schematic is determined when the sheet is created. For example, if the units are inches and the grid is 0.1, a sheet 22 inches wide is (22 / 0.1) or 220 Primary grids wide. In Secondary units, the sheet is 880 grids wide.

The ASCII to Schematic and Schematic to ASCII conversion utilities are called from the Utilities menu of the SCS Executive.

Interface Format

The interface is a line-oriented ASCII text format. Each element in the schematic is represented as a single line in the text file. Each line consists of:

- ♦ A keyword that begins the line and identifies the type of element described.
- ♦ A list of parameters describing the element following the keyword.

The following conventions apply within each line:

- ♦ Keywords are not case-sensitive.
- ♦ Parameters are delimited by spaces and tabs.
- ♦ Text strings appear at the end of the line and can contain embedded blanks.
- ♦ Coordinate locations are given as pairs of numbers. X coordinates increase to the right and Y coordinates increase to the bottom of the symbol. All coordinate locations and distances are given in Secondary grid units.

In the following description of the format, keywords are shown in roman text and parameters are shown in italics.

SCS Database Version

version *number*

The SCS symbol database version number is specified on the first line of the file.

Sheet Size

sheet *number width height*

The sheet entry should be the second line of the file. *number* is the sheet number and can range from 1 to 99. The *width* and *height* are the dimensions of the sheet description that follows. All other lines following the sheet entry are entities which appear on that sheet. Additional sheet entries represent the beginnings of new sheets.

Net Position (Wire)

```
wire x1 y1 x2 y2
```

The x-y coordinates are the wire's end points. These coordinates must lie on a major grid. These coordinates must also satisfy the interconnection constraints for wires. For example, they must be lines at 45 or 90 degree angles to the edge of the sheet.

Flag Position

```
flag x1 y1 name
```

The x-y coordinate is the location of the flag. This coordinate must lie on the Primary grid.

name is the net name to be displayed. A wire cannot display two different names. The *name* consists of the first non-blank character after *y1* and all following characters. The name must conform to the rules for net names. A net name cannot contain spaces.

Pin Position and Definition

```
I0pin x1 y1 in_out
```

The x-y coordinate is the location of the pin. This coordinate must lie on the Primary grid. *in_out* specifies the polarity and can be IN, OUT, or BIDIR. The first character can be used as an abbreviation.

Bus Tap Position

```
bustap x1 y1 x2 y2
```

The x-y coordinates are the end points of the wire forming a bus tap. These coordinates must lie on the Primary grid. The bus tap must be at a 90 degree angle to the bus and can be either horizontal or vertical. It must also satisfy the interconnection constraints. These constraints are listed on page 5-20 of "Using the Schematic Editor."

Symbol Position and Format

`symbol name x1 y1 rot_mir`

name is the name of the symbol. Since it is the name of a symbol file, it must conform to the file name conventions for SCS.

The *x-y* coordinate is the location of the symbol origin.

rot_mir specifies the rotation and mirroring applied to this symbol instance. *rot_mir* can have values of R0, R90, R180, R270, M0, M90, M180, or M270. The first two characters can be used as an abbreviation.

The R values represent the rotated positions of the object. The M values represent the rotated positions of the object after it has been mirrored left-right. The list below describes the orientation:

<i>rot_mir</i> Value	Orientation
R0	Default orientation, no rotation
R90	Default orientation, rotated 90° counter-clockwise
R180	Default orientation, rotated 180° counter-clockwise
R270	Default orientation, rotated 270° counter-clockwise
M0	Mirrored orientation, no rotation
M90	Mirrored orientation, rotated 90° counter-clockwise
M180	Mirrored orientation, rotated 180° counter-clockwise
M270	Mirrored orientation, rotated 270° counter-clockwise

`symattr name string`

The symbol instance attributes apply to the previously defined symbol instance.

name specifies the attribute and must match one of the attribute names listed in the Symbol Attributes section of the INI file.

string is the attribute value to be assigned. The *string* is the first non-blank character following the name field and all following characters.

Attribute Window Position

window number x1 y1 justify font

The attribute window location applies to the previously defined symbol instance. It overrides the window location in the symbol definition.

number specifies which attribute window is to be drawn.

The x-y coordinate is the origin for drawing the attribute value.

justify specifies the horizontal (vertical) justification of the text. It can have the value LEFT, RIGHT, CENTER, VLEFT, VRIGHT, or VCENTER. Any value can be abbreviated to just the first two characters. Horizontal (vertical) text is always justified vertically (horizontally) to the center of the character.

font specifies the size of the text and can be 0–7. (In the UNIX/Motif version, there are only three font sizes, 0–2. Sizes 4–7 are mapped to UNIX size 2, 3 is mapped to size 1, and 0–2 are mapped to size 0.)

Pin Attribute Position

pinattr x1 y1 name string

The x-y coordinate is the location of the pin. The specified pin attribute applies to the pin at this location.

name specifies the attribute and must match one of the attribute names listed in the Pin Attributes section of the INI file.

string is the attribute value to be assigned. The *string* is the first non-blank character following the name field and all following characters.

Net Attribute Position

netattr x1 y1 name string

The x-y coordinate is the location of a point on the net. The specified net attribute applies to the net at this location.

name specifies the attribute and must match one of the attribute names listed in the Net Attributes section of the INI file.

string is the attribute value to be assigned. The *string* is the first non-blank character following the name field and all following characters.

Each of the following formats for graphic elements contains a parameter that specifies line weight. This parameter can have the value NORMAL or WIDE. These keywords can be abbreviated as N or W.

On UNIX systems, the graphic-element format contains a parameter that specifies line style. This parameter can have the value Dash, Dot, or DashDot. If the parameter is missing, the line is drawn solid.

Lines `line width x1 y1 x2 y2 style`

The x-y coordinates are the line's end points.

Rectangles `rectangle width x1 y1 x2 y2 style`

The x-y coordinates are two opposite corners of the rectangle.

Circles `circle width x1 y1 x2 y2 style`

The x-y coordinates are the opposite corners of a square that encloses the circle. If a rectangle is specified, it is converted to a square with the same upper-left corner as the rectangle. The sides of the square are the length of the shorter sides of the rectangle.

Arcs `arc width x1 y1 x2 y2 x3 y3 x4 y4 style`

The first two x-y coordinates describe a circle (see above) containing the arc. The third and fourth coordinate pairs are the starting and ending points of the arc. If these points are not on the circle, the lines joining these points to the circle's center set the boundaries of the arc. The arc is drawn counterclockwise from the starting to the ending point.

Table Text Position

`text x1 y1 justify font string`

The x-y coordinate is the origin for drawing the text.

justify specifies the horizontal (vertical) justification of the text. It can have the value LEFT, RIGHT, CENTER, VLEFT, VRIGHT, or VCENTER. Any value can be abbreviated to just the first two characters. Horizontal (vertical) text is always justified vertically (horizontally) to the center of the character.

font specifies the size of the text and can be 0–7. (In the UNIX/Motif version, there are only three font sizes, 0–2. Sizes 4–7 are mapped to UNIX size 2, 3 is mapped to size 1, and 0–2 are mapped to size 0.)

string is the text string. The string consists of the first non-blank character after the *font* parameter and all following characters.

```
table x0 y0 rows cols row_height col_width
first_row_height first_col_width
name_just name_font title_just title_font
upper_left_just upper_left_font first_row_just first_row_font
first_col_just first_col_font rest_of_table_just
rest_of_table_font
```

The justification referred to below can have the following values:

```
BottomLeft
BottomCenter
BottomRight
CenterLeft
CenterCenter
CenterRight
TopLeft
TopCenter
TopRight
None
```

‘None’ is used if you don’t want the title or name to appear on the table.

The name and title are drawn outside the table. When the Bottom options are used to justify the table’s name or title, the table’s name is printed above the table. The Top options prints the table’s name below the table.

Font Parameters

The font parameters referred to below specify the text size and can be 0, 1, or 2. This corresponds to five, seven, or nine Secondary grid units in height.

Parameter	Description
<i>x0 y0</i>	The x-y coordinate is the upper-left corner of the table.
<i>rows</i>	Number of rows
<i>cols</i>	Number of columns
<i>row_height</i>	Height in Secondary grid units of all rows except the first

<i>col_width</i>	Width in Secondary grid units of all columns except the first
<i>first_row_height</i>	Height in Secondary grid units of the first row
<i>first_col_width</i>	Width in Secondary grid units of the first column
<i>title_just</i>	Justification for the table's title
<i>title_font</i>	Font size for the table's title
<i>name_just</i>	Justification for the table's name
<i>name_font</i>	Font size for the table's name
<i>upper_left_just</i>	Justification for the text placed in the upper-left cell
<i>upper_left_font</i>	Font size for the text placed in the upper-left cell
<i>first_row_just</i>	Justification for the table's first row (other than the first entry, covered by <i>upper_left_just</i>)
<i>first_row_font</i>	Font size for the table's first row (other than the first entry, covered by <i>upper_left_font</i>)
<i>first_col_just</i>	Justification for the table's first column (other than the first entry, covered by <i>upper_left_just</i>)
<i>name_font</i>	Font size for the table's first column (other than the first entry, covered by <i>upper_left_font</i>)
<i>rest_of_table_just</i>	Justification for all other cells, including all cells to the bottom right
<i>rest_of_table_font</i>	Font size for all other cells, including all cells to the bottom right

Table Name and Title Attributes

`tableattr number value`

The specified table attribute applies to the previously defined table.

number is the table attribute number. There are two choices:

0 Table name attribute

1 Table title attribute

where *value* is the value of the table attribute.

`tabledata row col value`

The table data refers to the previously defined table.

row and *col* specify the element in the table for which data is being defined.
value is the data.